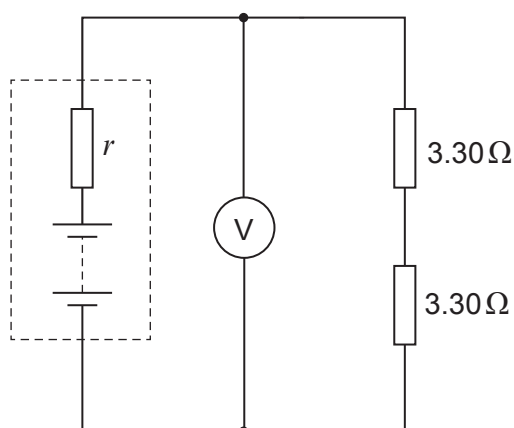


Answer all questions.

1. (a) Two $3.30\ \Omega$ resistors are connected in series across a battery of emf 4.80 V as shown. The voltmeter reads 4.33 V .

Diagram 1



- (i) State in terms of work or energy what is meant by the emf of a battery. [2]

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- (ii) Show that the internal resistance, r , of the battery is approximately $0.7\ \Omega$. [2]

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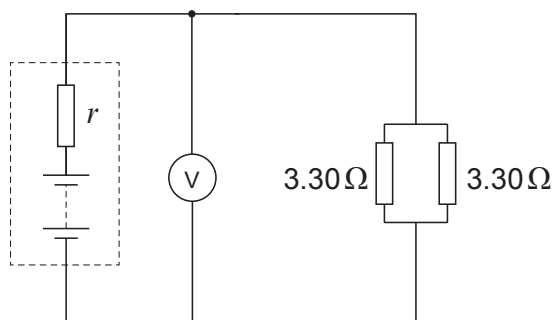
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- (iii) When the resistors are connected in parallel as shown below, the voltmeter reads 3.35 V .

Diagram 2



- (I) **Without further calculation** explain why you would expect the reading to be lower, now that there is a lower resistance across the cell terminals. [2]

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- (II) Calculate the number of electrons entering **either** of the resistors (shown in Diagram 2) per **minute**. [3]

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- (b) The heating element (a coil of wire) in an electric heater dissipates energy at a rate of 1.00 kW, when connected across the 230 V mains supply. Calculate:

- (i) the resistance of the coil; [2]

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- (ii) the energy dissipated per hour, giving your answer in megajoules (MJ). [1]

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- (c) For each megajoule of heat from an electric heater, approximately 0.08 m^3 of gas would have to be burned in a gas-fired electricity power station. For each megajoule of heat from a domestic gas fire or boiler, approximately 0.03 m^3 of gas is burned. Discuss whether the use of electric heaters in houses should be discouraged. Calculations are not required. [3]

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